

Executive Summary

Search and Rescue (S&R) and emergency response operations are frequently constrained by low-light and no-light conditions, where reduced visibility limits situational awareness, slows mobility, and increases risk to both responders and victims, especially in remote and rugged environments. As search efforts are highly time-sensitive, current solutions like handheld lights and vehicle-mounted systems can cause delays and inefficiency. The Aerial Matrix (Figure 1) addresses this gap by introducing a portable, aerial lighting and visual assistance system designed to provide dynamic, wide-area illumination while maintaining responder mobility.



Figure 1. Aerial Matrix System

The system is constrained by several key engineering and operational limitations that shape its design. A budget of \$1000 requires careful component selection while maintaining performance. Weight and payload constraints limit each drone to under 1 kg while supporting at least a 250 g payload, directly impacting material choice and hardware integration. Flight time must meet a minimum of 20 minutes to remain viable for field operations, placing restrictions on power consumption and battery capacity. Environmental conditions, including wind speeds up to 10 mph and light rain, require a durable and stable design. Additionally, reliable communication between drones over at least 5 meters with low latency is necessary to support coordinated operation, further constraining networking and control system design.

The Aerial Matrix system utilizes a coordinated network of lightweight drones equipped with modular lighting and visual sensing payloads to deliver adaptive illumination across complex terrain. Each drone operates within a synchronized flight system capable of maintaining positional accuracy while communicating through a mesh-based wireless network. Object detection enables the system to identify and track multiple targets in real time, reducing the need for continuous manual control. A user-friendly mobile interface allows operators to quickly deploy, configure, and monitor the system, providing intuitive control over drone positioning and targeting. The modular attachment design enables rapid payload changes, allowing the system to adapt to varying mission requirements such as S&R or infrastructure inspection.

By enhancing visibility and situational awareness without limiting operator mobility, the Aerial Matrix has the potential to significantly improve the speed and safety of S&R operations. Its scalable and modular architecture allows for future expansion, including increased swarm size, improved detection accuracy, and integration with additional sensing technologies. Beyond emergency response, the system may be applied to fields such as environmental monitoring, infrastructure inspection, and security operations. As autonomous aerial systems continue to evolve, the Aerial Matrix provides a foundation for more efficient, adaptable, and intelligent field-support technologies.